COLLEGE MANAGEMENT SYSTEM USING FILE HANDLING (PYTHON)

NAME : JAYAGURUNATHAN J

REG NO : 411625149016

SUBJECT : PYTHON

DEPARTMENT : CSE(CYBER SECUIRITY)

BATCH : 03

AIM

To develop a simple College Management System using Python that performs file handling operations such as adding, viewing, searching, and deleting student records.

PROCEDURE

1. Start the program.
2. Create a text file to store student records.
3. Define a function to add student details.
4. Accept roll number, name, and course from the user.
5. Write the data into the file using append mode.
6. Define a function to view all student records.
7. Read data from the file and display it.
8. Define a function to search for a student using roll number.
9. Define a function to delete a student record.

10. Create a menu-driven loop to perform operations until exit.

CODE

```
FILE_NAME = "students.txt"

def add_student():
    roll = input("Enter Roll Number: ")
    name = input("Enter Name: ")
    course = input("Enter Course: ")

    with open(FILE_NAME, "a") as file:
        file.write(roll + "," + name + "," + course + "\n")

    print("Student added successfully!\n")


def view_students():
    try:
        with open(FILE_NAME, "r") as file:
            data = file.readlines()
            if not data:
                print("No records found.\n")
                return

            print("\n--- Student Records ---")
            for line in data:
                roll, name, course = line.strip().split(",")
                print(f"Roll: {roll}, Name: {name}, Course: {course}")
            print()

    except FileNotFoundError:
        print("File not found.\n")


def search_student():
    search_roll = input("Enter Roll Number to search: ")
    found = False

    try:
        with open(FILE_NAME, "r") as file:
            for line in file:
                roll, name, course = line.strip().split(",")
                if roll == search_roll:
                    print(f"Found -> Name: {name}, Course: {course}\n")
                    found = True
```

```
            break

        if not found:
            print("Student not found.\n")

    except FileNotFoundError:
        print("File not found.\n")


def delete_student():
    delete_roll = input("Enter Roll Number to delete: ")
    lines = []

    try:
        with open(FILE_NAME, "r") as file:
            lines = file.readlines()

        with open(FILE_NAME, "w") as file:
            found = False
            for line in lines:
                roll, name, course = line.strip().split(",")
                if roll != delete_roll:
                    file.write(line)
                else:
                    found = True

        if found:
            print("Student deleted successfully!\n")
        else:
            print("Student not found.\n")

    except FileNotFoundError:
        print("File not found.\n")


while True:
    print("----- College Management System -----")
    print("1. Add Student")
    print("2. View Students")
    print("3. Search Student")
    print("4. Delete Student")
    print("5. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
```

```
        add_student()
    elif choice == "2":
        view_students()
    elif choice == "3":
        search_student()
    elif choice == "4":
        delete_student()
    elif choice == "5":
        print("Exiting program...")
        break
    else:
        print("Invalid choice! Try again.\n")
```

SAMPLE OUTPUT

```
----- College Management System -----
1. Add Student
2. View Students
3. Search Student
4. Delete Student
5. Exit

Enter your choice: 1
Enter Roll Number: 777
Enter Name: J GURU
Enter Course: BCA
Student added successfully!

Enter your choice: 2

--- Student Records ---
Roll: 777, Name: J GURU, Course: BCA
```

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main.py

```python
FILE_NAME = "students.txt"

def add_student():
    roll = input("Enter Roll Number: ")
    name = input("Enter Name: ")
    course = input("Enter Course: ")

    with open(FILE_NAME, "a") as file:
        file.write(roll + "," + name + "," + course + "\n")

    print("Student added successfully!\n")



def view_students():
    try:
        with open(FILE_NAME, "r") as file:
            data = file.readlines()
            if not data:
                print("No records found.\n")
                return

            print("\n--- Student Records ---")
            for line in data:
                roll, name, course = line.strip().split(",")
                print(f"Roll: {roll}, Name: {name}, Course: {course}")
            print()

    except FileNotFoundError:
        print("File not found.\n")



def search_student():
    search_roll = input("Enter Roll Number to search: ")
    found = False

    try:
        with open(FILE_NAME, "r") as file:
            for line in file:
                roll, name, course = line.strip().split(",")
                if roll == search_roll:
                    print(f"Found -> Name: {name}, Course: {course}\n")
                    found = True
                    break
```

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main.py

```python
                    found = True
                    break

        if not found:
            print("Student not found.\n")

    except FileNotFoundError:
        print("File not found.\n")



def delete_student():
    delete_roll = input("Enter Roll Number to delete: ")
    lines = []

    try:
        with open(FILE_NAME, "r") as file:
            lines = file.readlines()

        with open(FILE_NAME, "w") as file:
            found = False
            for line in lines:
                roll, name, course = line.strip().split(",")
                if roll != delete_roll:
                    file.write(line)
                else:
                    found = True

        if found:
            print("Student deleted successfully!\n")
        else:
            print("Student not found.\n")

    except FileNotFoundError:
        print("File not found.\n")


while True:
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```

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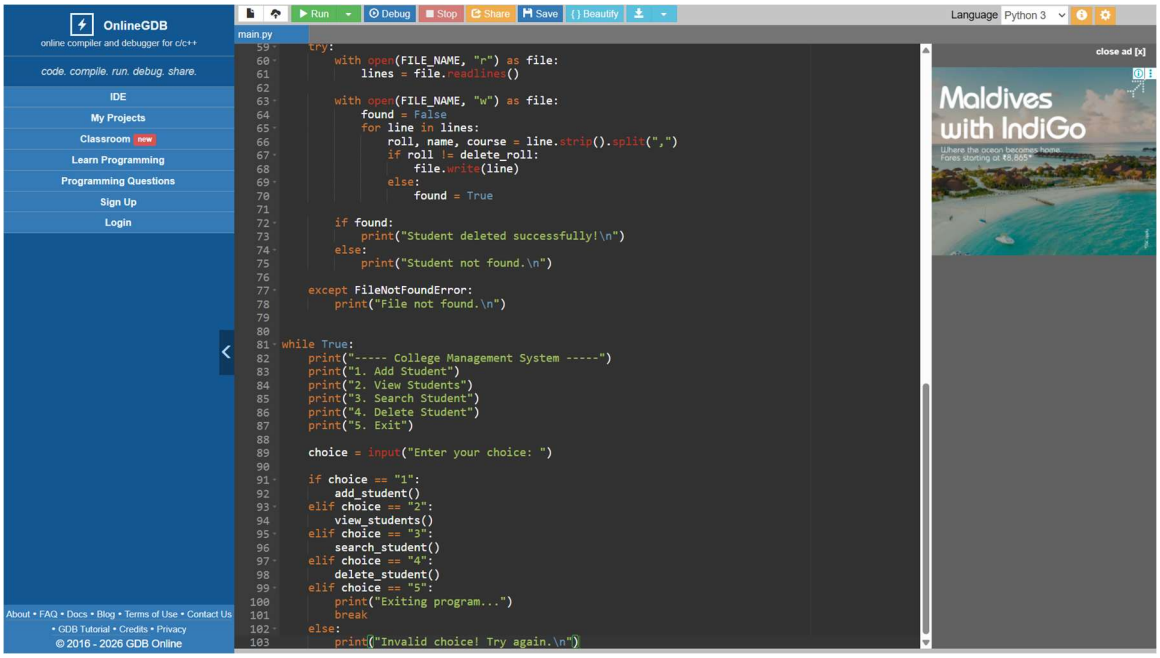


```
    try:
        with open(FILE_NAME, "r") as file:
            lines = file.readlines()

        with open(FILE_NAME, "w") as file:
            found = False
            for line in lines:
                roll, name, course = line.strip().split(",")
                if roll != delete_roll:
                    file.write(line)
                else:
                    found = True

        if found:
            print("Student deleted successfully!\n")
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while True:
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    choice = input("Enter your choice: ")

    if choice == "1":
        add_student()
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        view_students()
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        search_student()
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    elif choice == "5":
        print("Exiting program...")
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    else:
        print("Invalid choice! Try again.\n")
```

RESULT

Thus, the College Management System using Python was successfully implemented using file handling operations.